

Even and odd Dyck paths

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Abstract

Weight a Dyck path by the sum of the positions of its valleys. Among the paths of semilength n with k valleys, those of even weight outnumber those of odd weight, and Johann Cigler observed that the excess is exactly the number of symmetric paths with k valleys. He proved it by evaluating q -Narayana numbers at $q = -1$ and asked for a combinatorial proof. This note gives one. Valley coordinates turn each path into a Motzkin word in which the sign is a product of local signs; swapping letters inside pairs cancels every word except some of sign $+1$; and a halving map matches those survivors with the symmetric paths. The survivors are not the symmetric paths themselves, and cannot be: for odd n and k every symmetric path is odd. The argument has been checked in the Lean proof assistant.

1 Five paths

A *Dyck path* of semilength n is a sequence of n up steps U and n down steps D that never goes below its starting level. A *valley* is a down step followed at once by an up step; its *position* is the number of that down step, counting from 1. The *major index* $\text{maj } P$ is the sum of the positions of the valleys of P . Figure 1 shows the five paths of semilength 3.

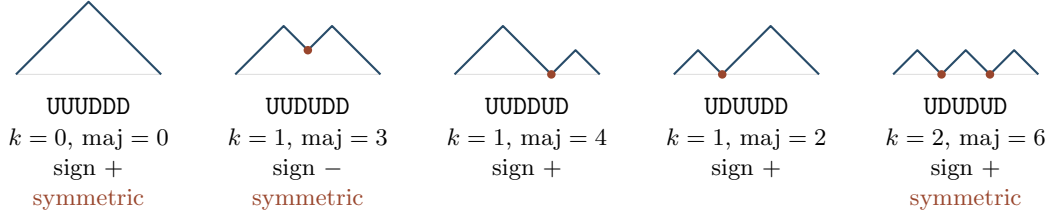


Figure 1: The Dyck paths of semilength 3. Valleys are marked; k is the number of valleys and the sign is $(-1)^{\text{maj}}$. A path is symmetric when it equals its mirror image, read backwards with U and D exchanged.

Stop and try this before reading on. Sort the paths of Figure 1 by their number of valleys. In each group, subtract the number of paths with odd major index from the number with even major index. Then count the symmetric paths in each group.

Both counts give 1, 1, 1. In the middle group the major indices are 3, 4 and 2, so the signed count is $1 + 1 - 1 = 1$, and there is one symmetric path. Write $E_{n,k}$ and $O_{n,k}$ for the numbers of paths of semilength n with k valleys and even or odd major index, and $S_{n,k}$ for the number of symmetric ones.

Theorem 1. For all $n \geq 1$ and $k \geq 0$, $E_{n,k} - O_{n,k} = S_{n,k}$.

Why should anyone care? The number of Dyck paths with k valleys is a Narayana number, and Förlinger and Hofbauer [4] found its natural q -analogue:

$$\sum_P q^{\text{maj } P} = q^{k(k+1)} \frac{1}{[k+1]_q} \binom{n}{k}_q \binom{n-1}{k}_q, \quad (1)$$

the sum running over the paths of semilength n with k valleys. The left side lists the paths by major index. The right side is the familiar product formula for the Narayana number, $\frac{1}{k+1} \binom{n}{k} \binom{n-1}{k}$, with each integer replaced by its q -analogue $[m]_q = 1 + q + \cdots + q^{m-1}$, and a power of q in front. Putting $q = -1$ on the left gives $E_{n,k} - O_{n,k}$, since $k(k+1)$ is even. Cigler [1] evaluated the right side at $q = -1$ and counted symmetric paths, finding they agree; on MathOverflow [3] he asked for a direct combinatorial proof, and Peter Taylor pointed out there that it amounts to Theorem 1. Identities of this shape, a q -count at $q = -1$ equal to the number of objects fixed by an involution, are instances of Stembridge's $q = -1$ phenomenon [6]. For even n the same value is also the half-turn case of a cyclic sieving theorem of Reiner, Stanton and White [5, Theorem 7.2], which counts noncrossing partitions; that proof, too, computes both sides. What follows is a direct argument for every n .

2 A tempting pairing that fails

The first idea is to pair each path with its mirror image P^* . It does not work, and seeing why shapes everything later.

A valley of P at position p consists of a down step at p and an up step at $p+1$. Reading backwards and exchanging the letters turns these into an up step at $2n+1-p$ and a down step at $2n-p$: a valley of P^* at position $2n-p$. So

$$\text{maj } P^* = 2nk - \text{maj } P,$$

which has the same parity as $\text{maj } P$. Mirror pairs never cancel. Worse, the valleys of a symmetric path come in pairs p and $2n-p$ (with $p = n$ unpaired), so a symmetric path has $\text{maj } P = nk$. When n and k are both odd, every symmetric path has odd major index and counts -1 on the left of Theorem 1, while it is meant to count $+1$ on the right. The path UUDUDD of Figure 1 is the smallest example.

So a cancellation argument has to throw away some symmetric paths and keep some that are not symmetric, and then match the survivors with the symmetric paths by a separate bijection. In Figure 1 the argument below cancels UUDUDD against UDUUDD and keeps UUDUDD.

3 Valleys as coordinates

The sign $(-1)^{\text{maj } P}$ depends on the positions of all the valleys at once. The first step is a change of description that makes it a product of local pieces.

Look at the path on the left of Figure 2. Its valleys are at positions 3 and 8. The valley at 3 has 2 up steps before it and is the 1st down step; the valley at 8 has 4 up steps before it and is the 4th down step. The positions are the sums: $3 = 2 + 1$ and $8 = 4 + 4$.

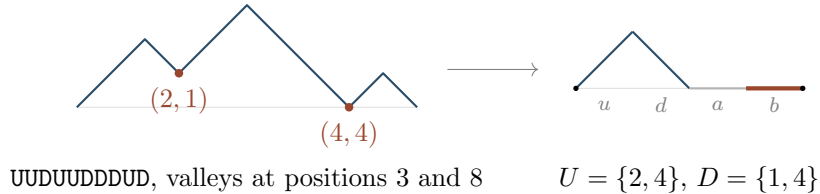


Figure 2: A path, its valley coordinates (u_j, d_j) , and its word. Reading $x = 1, 2, 3, 4$: x is in D only, in U only, in neither, in both, giving the letters u, d, a, b . The word is drawn as a path: u and d go up and down, a is a grey level step, b a thick rust one.

In general, for the j th valley let u_j be the number of up steps before it and d_j the number of down steps up to and including it. Its position is $u_j + d_j$, so

$$\text{maj } P = \sum_{j=1}^k u_j + \sum_{j=1}^k d_j. \quad (2)$$

In words, the major index splits into a contribution from the up steps and one from the down steps. The height of the path at the j th valley is $u_j - d_j$.

Lemma 1. *Let $n \geq 1$. The valley coordinates of a Dyck path of semilength n with k valleys satisfy*

$$1 \leq u_1 < \cdots < u_k \leq n-1, \quad 1 \leq d_1 < \cdots < d_k \leq n-1, \quad d_j \leq u_j \text{ for all } j,$$

and every such pair of sequences comes from exactly one Dyck path.

Proof. Between consecutive valleys a path makes one run of up steps and then one run of down steps, so $P = U^{u_1} D^{d_1} U^{u_2 - u_1} D^{d_2 - d_1} \cdots U^{n - u_k} D^{n - d_k}$. Every run is nonempty, which says that the sequences increase strictly and that u_k and d_k are less than n (after the last valley there are still up steps and down steps to come). Conversely this formula builds a sequence of n steps of each kind from any two such sequences. Its lowest points are its valleys and its end, so it stays at or above its starting level exactly when every valley height $u_j - d_j$ is at least 0. $\square$

Put $U = \{u_1, \dots, u_k\}$ and $D = \{d_1, \dots, d_k\}$, both subsets of $\{1, \dots, n-1\}$, and write a word w of length $n-1$ whose x th letter records whether x lies in D or U :

$$w_x = u \text{ if } x \in D \setminus U, \quad d \text{ if } x \in U \setminus D, \quad a \text{ if } x \notin U \cup D, \quad b \text{ if } x \in U \cap D.$$

Read u and d as up and down steps and a, b as two kinds of level step, as in Figure 2. The names are chosen so that the condition $d_j \leq u_j$ becomes a familiar one.

Lemma 2. *The words of Dyck paths of semilength n with k valleys are exactly the words of length $n-1$ over $\{u, d, a, b\}$ that never go below their starting level, end at it, and have k letters from $\{u, b\}$. Moreover*

$$(-1)^{\text{maj } P} = \prod_{x: w_x \in \{u, d\}} (-1)^x, \quad (3)$$

and P is symmetric exactly when w read backwards, with u and d exchanged, is w again.

Proof. After x letters the word has risen $|D \cap [1, x]| - |U \cap [1, x]|$, the number of d_j up to x minus the number of u_j up to x . If $d_j \leq u_j$ for all j , then each $u_j \leq x$ brings a $d_j \leq x$ with it, so this is never negative. Conversely, taking $x = u_j$ gives at least j values $d_i \leq u_j$, so $d_j \leq u_j$. The word ends at its starting level because $|U| = |D| = k$, and the letters from $\{u, b\}$ are the elements of D , so there are k of them. Lemma 1 turns these word conditions back into a unique path.

For (3), each x in exactly one of U and D appears once in (2), and each x in both appears twice, contributing $(-1)^{2x} = 1$.

For the mirror image, the valley at position $p_j = u_j + d_j$ becomes the valley at $2n - p_j$, which has $n - d_j$ up steps before it and is the $(n - u_j)$ th down step. So P^* has $U^* = n - D$ and $D^* = n - U$: its x th letter is the $(n - x)$ th letter of w with u and d exchanged. $\square$

Equation (3) is the point of the change. Each up or down letter carries the sign of its position and the level letters carry nothing. The five paths of Figure 1 have the words aa, ud, ab, ba and bb , with signs $+, -, +, +, +$.

4 Swapping inside pairs

By (3), moving a single u or d by one place changes the sign. That suggests cutting w into pairs of letters at positions $2i - 1$ and $2i$, leaving the last letter alone when $n - 1$ is odd, and looking for pairs that can be changed without disturbing anything else. Give some pairs a *partner* (Figure 3):

- a pair with exactly one level letter, such as ua or bd , is partnered with the same two letters in the other order;
- ud is partnered with ba ;
- du is partnered with ab , but only when the height before the pair is at least 1.

The pairs uu , dd , aa , bb have no partner, and neither has ab at height 0.

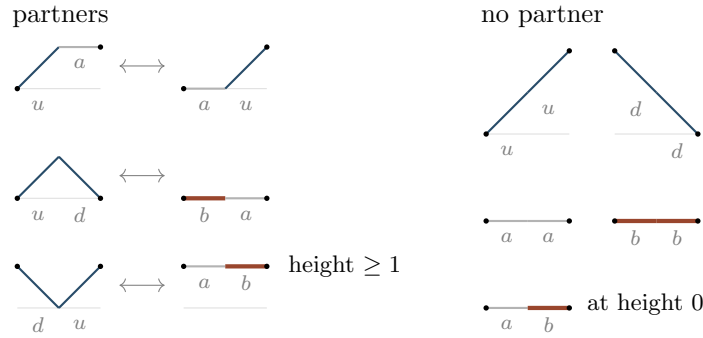


Figure 3: Partners. A swap moves one up or down letter by one place; ud and ba both go nowhere and both contain one letter from $\{u, b\}$; du dips, so it needs height 1 to stand on.

The rule is: find the first pair in w that has a partner, and replace it by its partner. For the word $udab$ of Figure 2 the first pair is ud , so the rule gives $baab$, the word of UDUUDDDDUD. The major indices are 11 and 10.

Lemma 3. *The rule is an involution on the words of Lemma 2 with k letters from $\{u, b\}$. When it moves a word, it changes its sign. A word it fixes has sign $+1$.*

Proof. A pair and its partner rise by the same amount, so the heights before all later pairs are unchanged; the earlier pairs have no partner and are untouched. Applying the rule again therefore finds the same pair and puts it back. The new word is legal, because a swap changes only the height in the middle of the pair. That middle height is the height before the pair when a level letter comes first, the height after the pair when the level letter comes second, and one more than the height before the pair for ud ; all three are at least 0 in a legal word. The remaining case is du , whose middle height is one less than the height before it, and that is what the condition on its partner allows for. The number of letters from $\{u, b\}$ is kept, since ud and ba each have one, as do du and ab .

For the sign, use (3). A swap moves one up or down letter between positions $2i - 1$ and $2i$, which flips its sign. The pairs ud and du contribute $(-1)^{2i-1}(-1)^{2i} = -1$, and the pairs ba and ab contribute $+1$.

A fixed word consists of pairs uu , dd , aa , bb , and ab at height 0, followed perhaps by a lone last letter. The lone letter is a level letter, because the pairs use an even number of up and down letters and a word that returns to its start has as many u as d , hence an even number of them in all. Each uu and each dd contributes -1 to the sign, and there are as many of one as of the other, because no other pair changes the height. So the sign is $+1$. $\square$

Summing $(-1)^{\text{maj } P}$ over all paths with k valleys, the words moved by the rule cancel in pairs, and so

$$E_{n,k} - O_{n,k} = \text{the number of fixed words with } k \text{ letters from } \{u, b\}. \quad (4)$$

For $n = 3$ the words are single pairs: ud and ba cancel, and the fixed words are aa , ab and bb , one in each group, as in Figure 1.

5 From fixed words to symmetric paths

The last step matches fixed words with mirror-symmetric words. Take the fixed word $abuu dd ab$ at the top of Figure 4. Each of its pairs changes the height by 0 or ± 2 , so the heights between pairs are even, and it can be halved: write one letter per pair, $uu \mapsto u$, $dd \mapsto d$, $aa \mapsto a$, $bb \mapsto b$, and $ab \mapsto c$, a new level letter. The result $cudc$ returns to its start and has c only at height 0. Now turn each c into u , giving $uudu$. This word never goes below its start, but it no longer needs to end there. The steps that were c have a description of their own: the first is the last step leaving level 0, the second the last step leaving level 1. Finally append the mirror image $dudd$, to get the symmetric word $uudu dudd$.

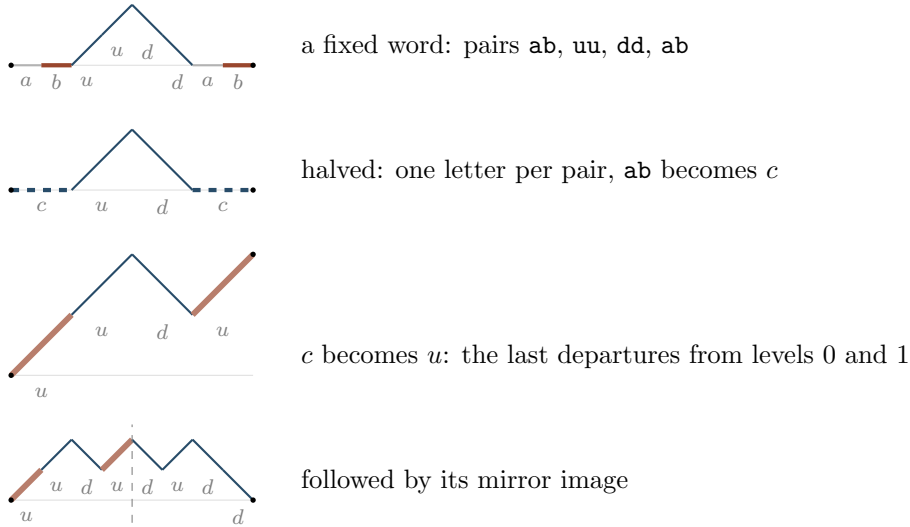


Figure 4: Halving a fixed word, turning c into u , and adding the mirror image. The two highlighted steps are the last departures from levels 0 and 1; they are where the c letters were.

Lemma 4. *Halving, turning c into u , keeping the lone last letter (if any) as a middle letter, and appending the mirror image, is a bijection from the fixed words of length $n - 1$ to the words of Lemma 2 that equal their mirror image. It keeps the number of letters from $\{u, b\}$.*

Proof. Write h for the word obtained after turning c into u . Each c occurs at height 0 of the halved word, and turning the j th one into u raises everything after it by one more, so at that step the new word climbs from $j - 1$ to j and afterwards stays at height at least j . The former c letters are therefore the last departures of h from the levels $0, 1, \dots, e - 1$, where e is the final height of h . Conversely, in any word h that never goes below its start and ends at height e , marking the last departure from each level $0, \dots, e - 1$ and turning those letters into c lowers the height after the j th mark by j . Before the j th mark the word sits at height at least $j - 1$, so the lowered word never goes below 0, it is at height 0 at every c , and it ends at 0. Doubling each letter back ($c \mapsto ab$) recovers the fixed word.

A word equal to its mirror image is h , then a middle letter when $n - 1$ is odd (which the mirror condition forces to be a or b), then the mirror of h ; it never goes below its start exactly when h does not. So the map is a bijection.

Count letters from $\{u, b\}$. In the fixed word there are $2\#uu + 2\#bb + \#ab$ of them, plus one for a final b . In the symmetric word, the mirror of h has as many u as h has d , so there are $\#u(h) + \#d(h) + 2\#b(h)$, plus one for a middle b . Here $\#u(h) = \#uu + \#ab$, $\#d(h) = \#dd = \#uu$ and $\#b(h) = \#bb$, and the two counts agree. $\square$

Proof of Theorem 1. By (4), $E_{n,k} - O_{n,k}$ is the number of fixed words with k letters from $\{u, b\}$. By Lemma 4 this is the number of mirror-symmetric words with k such letters, and by Lemma 2 these are the words of the symmetric Dyck paths with k valleys. $\square$

In Figure 1, the fixed word ab of $UUDDUD$ halves to c , becomes u , and with its mirror gives ud , the word of the symmetric path $UUDUDD$.

6 Remarks

The proof uses only the parity of the major index, together with the identity (1) of Förlinger and Hofbauer, which is quoted and not proved here. The cancellation is the same device, swapping inside pairs of steps, that Martin Rubey used to evaluate weighted Motzkin paths in a proof reported by Cigler [2, proof of the Lemma in Section 5]; the new ingredients are the valley coordinates, which make the sign of a Dyck path local, and the last-departure bijection.

The involution and the bijection do not commute with taking mirror images, and Section 2 shows that no cancellation can fix exactly the symmetric paths. A single map from the even paths to the odd paths together with the symmetric ones, which Taylor’s reformulation also invites, is not constructed here.

Verification. The whole argument, from Dyck paths written as lists of steps to Theorem 1, is formalised in Lean 4 with Mathlib [7], using only the standard axioms; identity (1) is the one statement taken from the literature. A Python program checks every claim of Sections 3 to 5 for all paths with $n \leq 12$. An independent program, sharing no code with it, recomputes the examples in the text, confirms Theorem 1 and its agreement with the right side of (1) at $q = -1$ for $n \leq 10$, and checks (1) itself coefficient by coefficient for $n \leq 8$. Both, with the Lean files, are at <https://github.com/jvvk/mathematics/tree/main/even-and-odd-dyck-paths>.

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